

Delivery Experience Optimization

Product Analytics & MVP Proposal

"Customers don't necessarily need faster deliveries — they need more predictable deliveries and better communication when something changes."

AUTHOR	ROLE	DATASET	APPLICATION
Nardos Yinges	Area Manager, Delivery Station	5,000 Synthetic Orders	Google APM Port

19.3%

LATE DELIVERY
RATE

3.84

AVG
SATISFACTION /5.0

12.4%

SUPPORT
CONTACT RATE

70.0%

REPEAT PURCHASE
RATE

86.0%

AVG ETA
ACCURACY

Executive Summary

Key findings from 5,000 synthetic delivery orders

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19.3%

LATE
DELIVERY
RATE↑ Primary pain
point

3.84

AVG
SATISFACTION
/5.0Drops to 2.72
when late

12.4%

SUPPORT
CONTACT RATE

39.3% when late

70.0%

REPEAT
PURCHASE
RATEFalls to 45.8%
when late

86.0%

AVG ETA
ACCURACYStrongest
predictor

CORE FINDING

ETA accuracy is the **strongest predictor** of customer satisfaction, support contacts, and repeat purchases. Moving a customer from the 70–80% ETA accuracy bracket to 90–100% improves satisfaction by **+0.62 points**, reduces support contacts by **–10.5 pp**, and increases repeat purchases by **+13.6 pp**.

The Problem: Late Delivery Cascade

How late deliveries degrade every customer metric

Metric	On-Time	Late	Delta
Satisfaction Score	4.11	2.72	–1.39
Support Contact Rate	6.0%	39.3%	+33.3 pp
Repeat Purchase Rate	75.8%	45.8%	–30.0 pp
% of Total Orders	80.7%	19.3%	—

The Opportunity: Notification Impact

Among late deliveries only (n = 967)

Metric	No Notif.	Notified	Delta
Satisfaction	2.56	3.12	+0.57
Support Rate	42.2%	32.3%	–9.9 pp
Repeat Purchase	41.9%	55.3%	+13.4 pp
Order Count	685	282	30% treated

PRODUCT RECOMMENDATION

Build **Proactive Delivery Status Notifications** — detect at-risk orders and communicate updated ETAs before customers discover the delay. Expected impact: **–10 pp support contacts**, **+0.57 satisfaction points**, **+13.4 pp repeat purchases**. Design an A/B test to validate at 50K customers per group over 4 weeks.

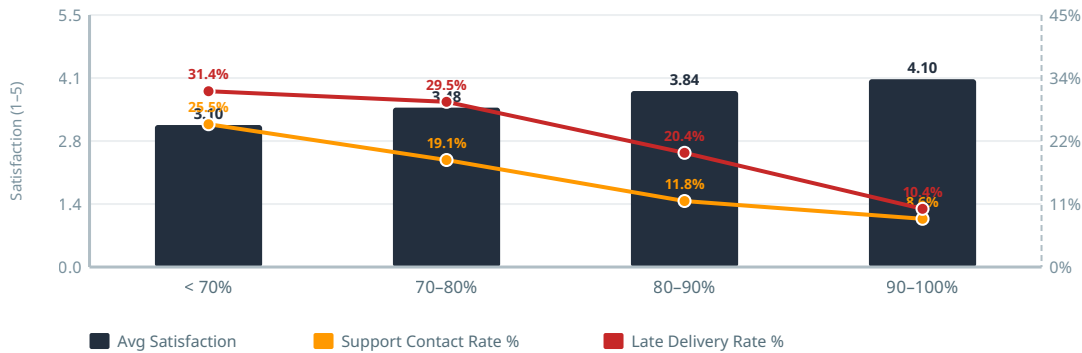
ETA Accuracy Drives Customer Experience

The strongest predictor across all customer outcome metrics

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ETA Accuracy vs. Satisfaction, Support Rate & Late Rate

Bars = Avg Satisfaction (left axis, 1–5) • Orange line = Support Contact Rate % (right axis) • Red line = Late Delivery Rate % (right axis)



ETA Accuracy Breakdown Table

ETA Accuracy	Orders	Late Rate	Satisfaction	Support Rate	Repeat Purchase
< 70%	51	31.4%	3.10	25.5%	45.1%
70-80%	931	29.5%	3.48	19.1%	61.4%
80-90%	2,594	20.4%	3.84	11.8%	70.7%
90-100%	1,424	10.4%	4.10	8.6%	75.0%

KEY DELTA

Moving 70-80% → 90-100%
ETA accuracy:
+0.62 satisfaction points
-10.5 pp support contacts
+13.6 pp repeat purchases

OPERATIONAL LINK

As an Area Manager, signals that predict ETA deviation — route density, driver pace, weather — are visible in-system **30-60 min before** customers notice.

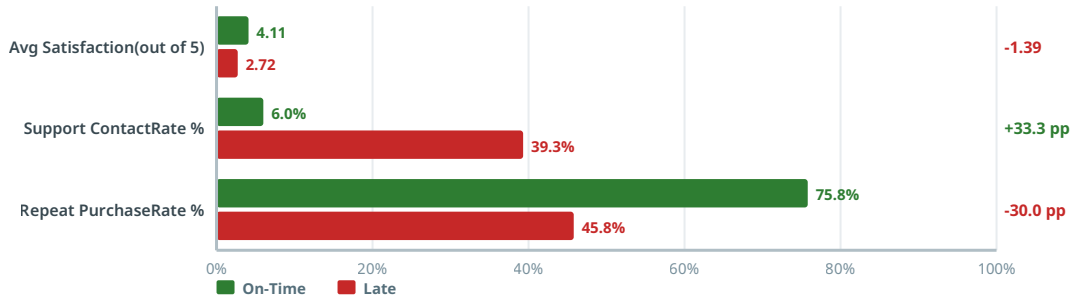
Late Deliveries Devastate Customer Metrics

19.3% of orders are late — their impact is disproportionate

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On-Time vs. Late Delivery — Customer Outcome Comparison

■ Green = On-Time • ■ Red = Late • Deltas shown on right axis



6.5×

MORE LIKELY TO
CONTACT SUPPORT

6.0% → 39.3%

−1.39

SATISFACTION
SCORE DROP

4.11 → 2.72 / 5.0

−30 pp

REPEAT PURCHASE
RATE DROP

75.8% → 45.8%

967

LATE ORDERS IN
DATASET

19.3% of 5,000 orders

■ WHY THIS MATTERS

A single late delivery doesn't just affect the customer's satisfaction score — it reduces the probability they order again by **30 percentage points**. At scale, every percentage point improvement in late rate directly recovers customer lifetime value.

■ THE GAP

Late deliveries are often operationally unavoidable. But the **customer experience** of a late delivery can be significantly improved — without fixing the underlying delivery. This is the product opportunity.

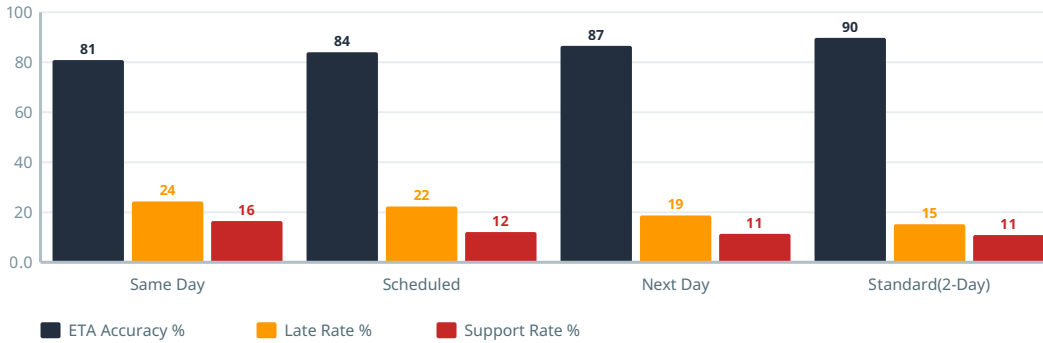
Performance by Delivery Type

Same Day carries the highest operational and customer risk

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ETA Accuracy, Late Rate & Support Rate by Delivery Type

■ Navy = ETA Accuracy % • Orange = Late Rate % • Red = Support Contact Rate %



Delivery Type Performance Summary

Delivery Type	Orders	ETA Accuracy	Late Rate	Satisfaction	Support Rate
Same Day	1,014	80.9%	24.3%	3.65	16.5%
Scheduled	718	84.0%	22.3%	3.78	12.1%
Next Day	1,844	86.6%	18.7%	3.88	11.4%
Standard (2-Day)	1,424	89.8%	15.2%	3.96	10.9%

PRIORITY TARGET

Same Day has the worst profile across all metrics — lowest ETA accuracy (80.9%), highest late rate (24.3%), and lowest satisfaction (3.65). This is the **highest-impact segment** for proactive notification rollout.

OPERATIONAL NOTE

Same Day's tight delivery windows leave less buffer for route variability — a known challenge at the delivery station level. The system has the data to predict these failures early.

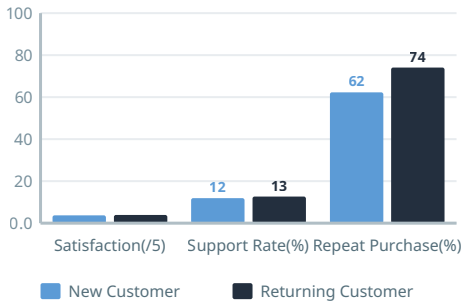
Customer Segment Analysis

New vs. Returning customers and ETA impact on retention

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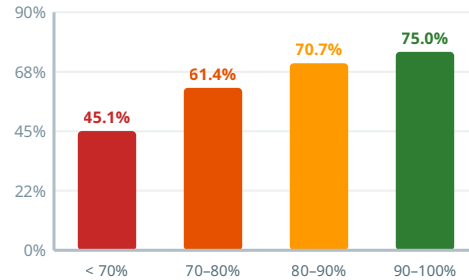
New vs. Returning Customer Outcomes

■ New (n=1,771) • ■ Returning (n=3,229)



ETA Accuracy → Repeat Purchase Rate

Higher ETA accuracy directly drives customer retention



New vs. Returning — Full Breakdown

Customer Type	Orders	Satisfaction	Support Rate	Repeat Purchase
New	1,771	3.76	11.9%	62.3%
Returning	3,229	3.88	12.7%	74.1%

- Returning customers show higher baseline satisfaction (+0.12) and significantly better repeat purchase rates (+11.8 pp).
- New customers are more vulnerable to churn from a bad first delivery.

NEW CUSTOMER RISK

New customers have a **62.3%** repeat purchase rate vs **74.1%** for returning. A bad first delivery — late, no notification

dramatically increases churn risk.

Proactive notifications are most critical for this segment.

Customers in the 90-100% ETA accuracy bracket have a **75.0%** repeat purchase rate. The path from 45.1%

(worst ETA) to 75.0% (best ETA) is **+29.9**

pp — the product opportunity.

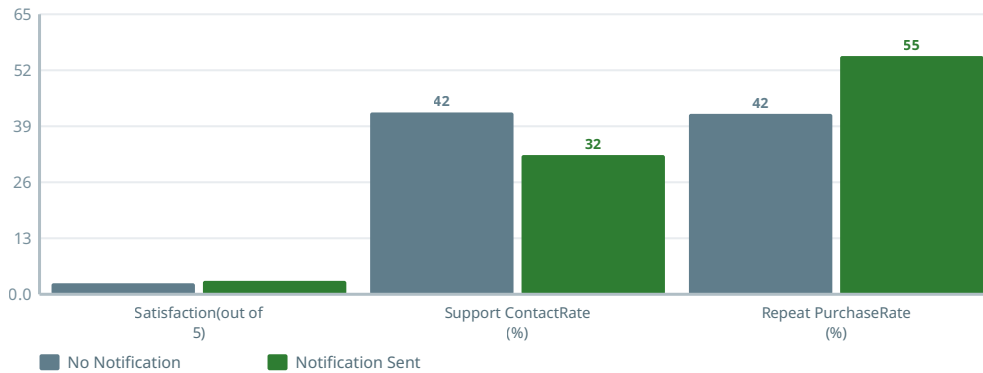
Proactive Notifications Fix the Customer Experience

Among 967 late deliveries — notification impact on all key metrics

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★ The Money Chart: Notification Impact on Late Deliveries

■ Gray = No Notification (n=685) • ■ Green = Notification Sent (n=282) • Late deliveries only



+0.57

SATISFACTION IMPROVEMENT

2.56 → 3.12 / 5.0

−9.9 pp

SUPPORT CONTACT
REDUCTION

42.2% → 32.3%

+13.4 pp

REPEAT PURCHASE INCREASE

41.9% → 55.3%

“Proactive notifications don’t fix the delivery — they fix the customer experience.”

CORE PRODUCT THESIS • DELIVERY EXPERIENCE OPTIMIZATION

■ WHY IT WORKS

Customers don’t react to *lateness* — they react to *uncertainty*. When they know what to expect and when, the emotional experience of a late delivery is fundamentally different. **Information replaces frustration with agency.**

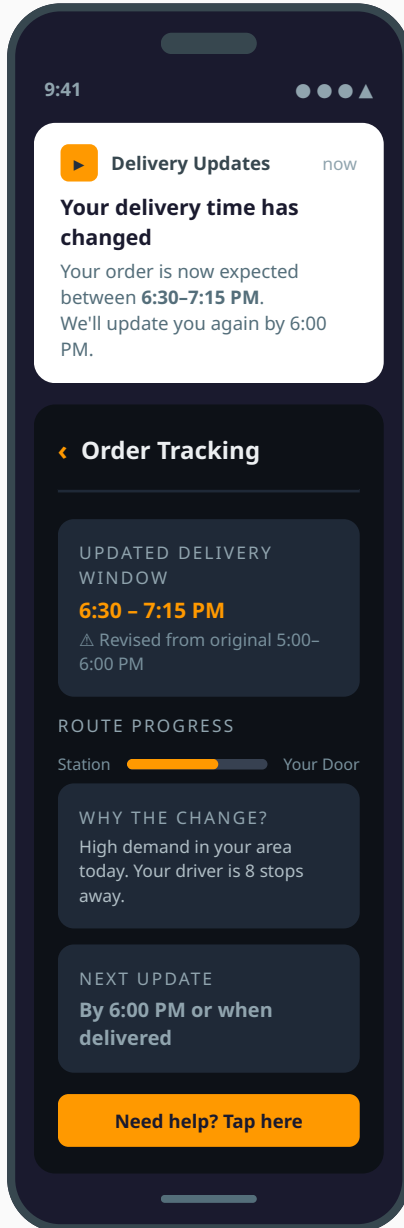
■ BUSINESS CASE

At 967 late orders per 5,000 (19.3%), if only 30% currently receive notifications, scaling to 100% notification coverage would save **~67 support contacts** and recover **~130 repeat purchases** per 5,000 orders.

Proposed MVP: Proactive Delivery Status

When at-risk orders are detected, customers are informed before they find out themselves

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Mobile notification UI mockup

MVP REQUIREMENTS

MUST HAVE P0

- ▶ At-risk order detection (>60% probability of missing ETA window)
- ▶ Updated ETA calculation (time range, not point estimate)
- ▶ Proactive push notification within 2 min of detection
- ▶ Clear explanation of what changed
- ▶ Next update time commitment
- ▶ Updated in-app order tracking page

FUTURE P2

- ▶ Personalized notification preferences
- ▶ One-tap support from notification
- ▶ Delivery rescheduling from notification
- ▶ Intelligent explanations (weather, volume)
- ▶ Proactive compensation for major delays

USER STORY

"As a customer waiting for a delivery, I want to know as early as possible when my delivery is at risk of being late, so I can adjust my plans and feel informed rather than frustrated."

TECHNICAL REQUIREMENTS

Detection latency: Within 5 min of signal change • **Notification delivery:** Within 2 min of detection • **ETA recalculation:** Within 30 seconds • **Min lead time:** 15 min before original ETA

A/B Test: Proactive Delivery Status

Randomized controlled test to validate product impact before full rollout

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CONTROL GROUP

Experience **Current delivery tracking**

Notifications **None when at-risk**

Customer discovers delay **When ETA window passes**

Sample size **50,000 customers**

Randomization **Customer-level**

TREATMENT GROUP

Experience **Proactive Delivery Status**

Trigger **>60% probability of missing ETA**

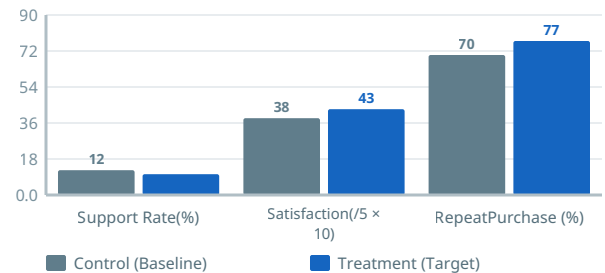
Notification **Updated ETA + next update time**

Cap **Max 2 notifications per order**

Duration **4 weeks (28 days)**

Expected Outcome vs. Baseline

Sat. scaled $\times 10$ for visibility



Metrics Framework

Type	Metric	Target
Primary	Support contact rate (at-risk orders)	-10 pp
Secondary	Customer satisfaction (CSAT)	+0.5 pts
Secondary	Repeat purchase rate (30-day)	+10 pp
Secondary	Notification engagement (open rate)	Measure
Guardrail	Order cancellation rate	< +1 pp
Guardrail	Notification opt-out rate	< 5%
Guardrail	False positive notification rate	< 15%

Decision Framework

Outcome	Action
Primary improves + no guardrail violations	► Ship to 100%
Primary improves, borderline guardrail	► Extend 2 weeks
Primary neutral, secondary improves	► Iterate & re-test
Primary worsens / guardrail violated	► Stop + RCA

SAMPLE SIZE RATIONALE

Based on baseline support rate 12.4%, MDE of 2pp, 80% power, $\alpha=0.05 \rightarrow$ **50K customers/group**, 100K total.

From Operations to Product

How Nardos's background maps to each phase of this project

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Experience	Connection to This Project
Area Manager, Delivery Station	Operational understanding of last-mile variability — route density, driver pace, volume spikes. The signals that predict late deliveries are visible in-system 30–60 min before customers notice.
300+ Associate Management	Stakeholder coordination at scale. Communication is the difference between a frustrated team and an aligned one — the same principle applies to customers.
Cortex / SCC / PerfectMile	Real-time monitoring experience informs the at-risk detection model — the data inputs required for proactive notification already exist in operational tools.
Data Analytics Coursework (UW)	Python analysis, statistical reasoning, and data-driven decision making — applied directly to generating and analyzing the 5,000-order synthetic dataset.
Business Degree + TechFix PM	Customer-centric thinking, financial impact modeling, and technical team coordination — all reflected in the PRD structure and experiment design.

REPOSITORY STRUCTURE

- ▶ **delivery_experience.csv** — 5,000 synthetic orders, 16 features
- ▶ **analysis.py** — Full Python analysis script
- ▶ **PRD.md** — Product Requirements Document
- ▶ **user_stories.md** — 8 user stories with acceptance criteria
- ▶ **experiment_plan.md** — Full A/B test design

TOOLS & TECHNOLOGIES

- ▶ **Python** (pandas, numpy) — data generation & analysis
- ▶ **Markdown** — PRD, user stories, experiment plan
- ▶ **PDF Dashboard** — this document (portfolio artifact)
- ▶ **Git/GitHub** — version control & portfolio presentation
- ▶ **Synthetic dataset** — no proprietary data used